Game Theory for Data Science

Midterm Exam • Fall 2023

Instructor: Dr. S. Iyer • October 19, 2023 • 75 minutes • 60 points

Name: _____ Student ID: _____ Section: _____

Read these instructions first.

- This exam has three parts. **Part A** (18 pts): six true/false questions; circle T or F *and* give a one- or two-sentence justification (a correct T/F with no justification earns 1 of 3 points). **Part B** (18 pts): six multiple-choice questions with exactly one correct answer each; no justification is required. **Part C** (24 pts): two short problems; show your work.
- Closed book. One page (one side) of notes is allowed. Calculators are allowed but not needed; $\sqrt{3} \approx 1.732$.
- Write your answers in the space provided. Use the back of the page if you need more room and say so.

Part	A	B	C	Total
Points	18	18	24	60
Score				

Part A: True or False (6 × 3 = 18 points)

A1. Every finite two-player game has at least one Nash equilibrium in pure strategies. **T / F**

A2. In a finite extensive-form game of perfect information, every Nash equilibrium is subgame perfect. **T / F**

A3. In a finite two-player zero-sum game, if (x, y) and (x', y') are both (mixed) Nash equilibria, then (x, y') is also a Nash equilibrium. **T / F**

A4. If (x, y) is a mixed-strategy Nash equilibrium of a two-player game, then the product distribution $p(i, j) = x_i \, y_j$ over action profiles is a correlated equilibrium. **T / F**

A5. In a finite game with an exact potential function Φ (so that every unilateral deviation changes the deviator's payoff by exactly the change in Φ), every pure profile that maximizes Φ is a pure Nash equilibrium. **T / F**

A6. In every finite two-player game, the empirical frequencies of play generated by fictitious play converge to a Nash equilibrium. **T / F**

Part B: Multiple Choice (6 × 3 = 18 points)

B1. Two data teams can pool their effort on a shared model (*Stag*) or each fine-tune their own (*Hare*). Payoffs:

	Stag	Hare
Stag	6, 6	0, 4
Hare	4, 0	3, 3

In the (fully) mixed Nash equilibrium, each team plays Stag with probability

(a) $1/2$ (b) $2/5$ (c) $3/5$ (d) $3/4$ (e) there is no fully mixed equilibrium

B2. Two autonomous vehicles meet at an intersection and choose *Go* or *Stop*:

	Go	Stop
Go	$-10, -10$	1, 0
Stop	0, 1	0, 0

Consider the distributions over (Row, Column) action profiles:
I: $\frac{1}{2}$ on (Go, Stop), $\frac{1}{2}$ on (Stop, Go).
II: $\frac{1}{3}$ each on (Go, Stop), (Stop, Go), (Stop, Stop).
III: $\frac{1}{4}$ on each of the four profiles.
Which are correlated equilibria?

(a) I only (b) I and II only (c) I and III only (d) I, II and III (e) none of them

B3. A learner runs the multiplicative-weights (Hedge) algorithm over n experts for T rounds with losses in $[0, 1]$ chosen by an adversary, using the best fixed learning rate for the known T. Its expected regret with respect to the best single expert grows as

(a) $O(1)$ (b) $O(\log T)$ (c) $O(\sqrt{T \ln n})$ (d) $O(T / \ln n)$ (e) $O(n\sqrt{T})$

B4. In *nonatomic* selfish routing where every edge has an affine latency function $\ell_e(x) = a_e x + b_e$ with $a_e, b_e \geq 0$, the largest possible price of anarchy (over all networks and demands) is

(a) 1 (b) $4/3$ (c) 2 (d) $5/2$ (e) unbounded

B5. Which of the following games is *not* an exact potential game? (Row player's payoff first.)

(a)

$3,3$	$0,5$
$5,0$	$1,1$

(b)

$3,2$	$0,0$
$0,0$	$2,3$

(c)

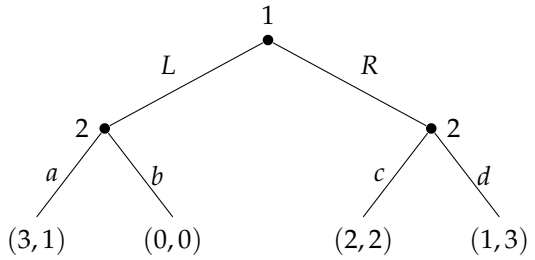
$2,2$	$4,4$
$4,4$	$2,2$

(d)

$1,-1$	$-1,1$
$-1,1$	$1,-1$

(e) All four are exact potential games.

B6. In the game tree below, a strategy for player 2 is written as (action after L, action after R).



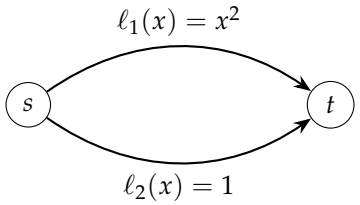
Which profile is a Nash equilibrium that is *not* subgame perfect?

(a) $(L, (a, d))$ (b) $(R, (b, d))$ (c) $(R, (b, c))$ (d) $(L, (b, d))$ (e) none of these

Part C: Problems (24 points)

C1. A steep Pigou network (12 points)

One unit of traffic (a continuum of infinitesimal drivers) travels from s to t over two parallel links. The top link has latency $\ell_1(x) = x^2$ when it carries flow x; the bottom link has constant latency $\ell_2(x) = 1$.



(a) Find the Wardrop (Nash) equilibrium flow and the total travel time. (3 pts)

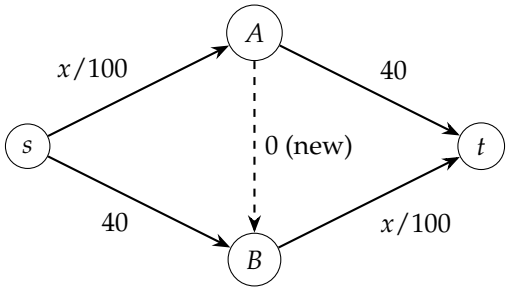
(b) Find the flow that minimizes total travel time $C(x) = x\,\ell_1(x) + (1-x)\,\ell_2(1-x)$, and the minimum total travel time. (4 pts)

(c) Compute the price of anarchy. How does it compare with the value for the classic Pigou network with $\ell_1(x) = x$? (2 pts)

(d) A toll τ is charged on the top link, and drivers minimize latency plus toll. Which toll makes the optimal flow from (b) an equilibrium? (3 pts)

C2. Braess's paradox in a delivery network (12 points)

Every morning 3000 delivery vans travel from depot s to hub t. The network is shown below. Edges $s \to A$ and $B \to t$ are congestible, with travel time $x/100$ minutes when x vans use them; edges $A \to t$ and $s \to B$ take 40 minutes regardless of traffic. Each van chooses a path to minimize its own travel time (treat vans as a continuum).



(a) Without the dashed edge $A \to B$, find the equilibrium and each van's travel time. (2 pts)

(b) A zero-travel-time connector $A \to B$ is opened. Show that in equilibrium every van uses the path $s \to A \to B \to t$, and find the new travel time. (4 pts)

(c) With the connector open, find the routing that minimizes total travel time, the minimum total travel time, and the price of anarchy. (4 pts)

(d) Did the connector help or hurt? Explain the difference between its effect on the equilibrium and on the optimum. (2 pts)

End of exam. Check that you have answered every question.